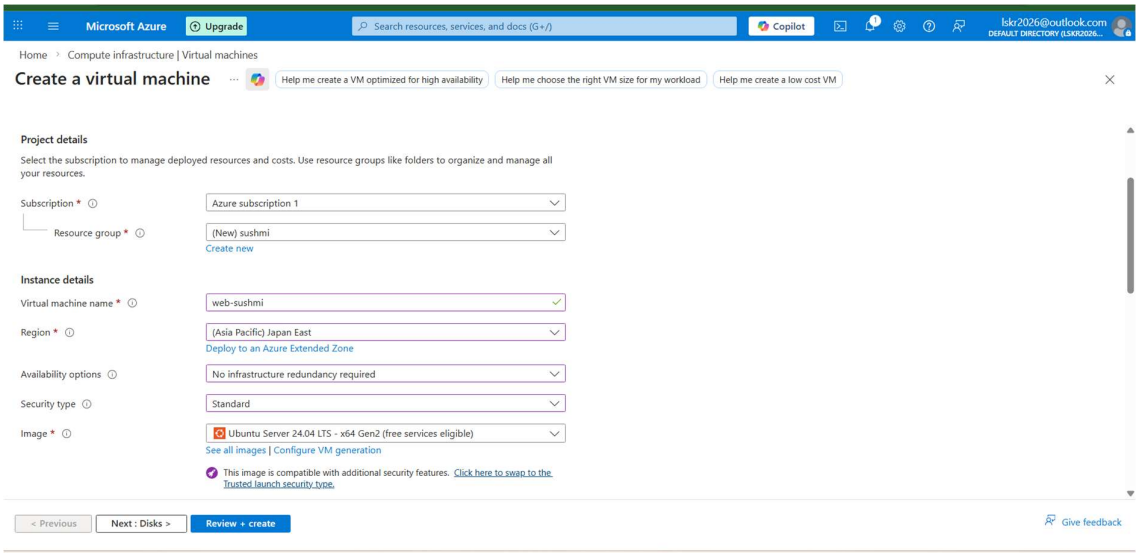
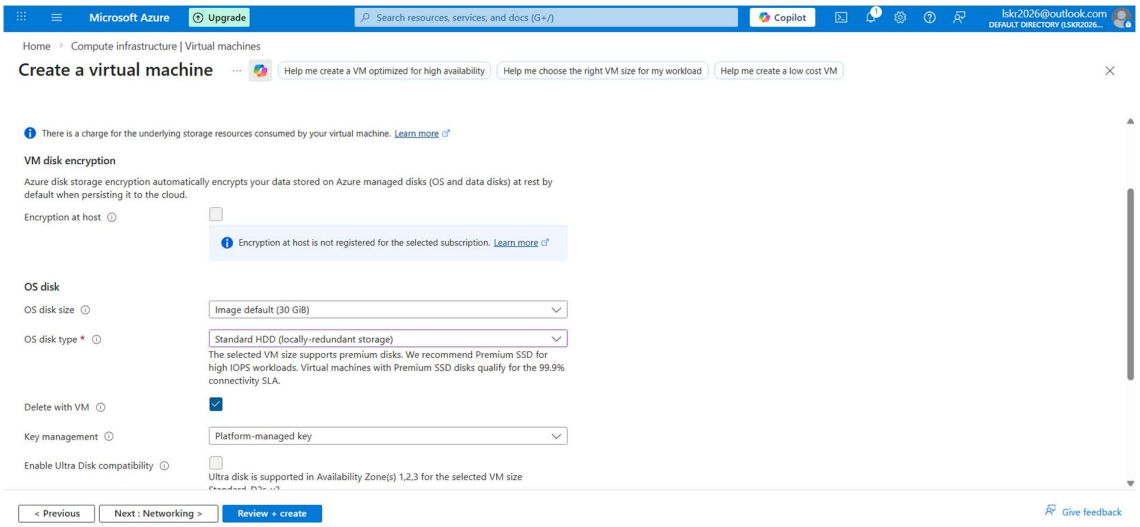
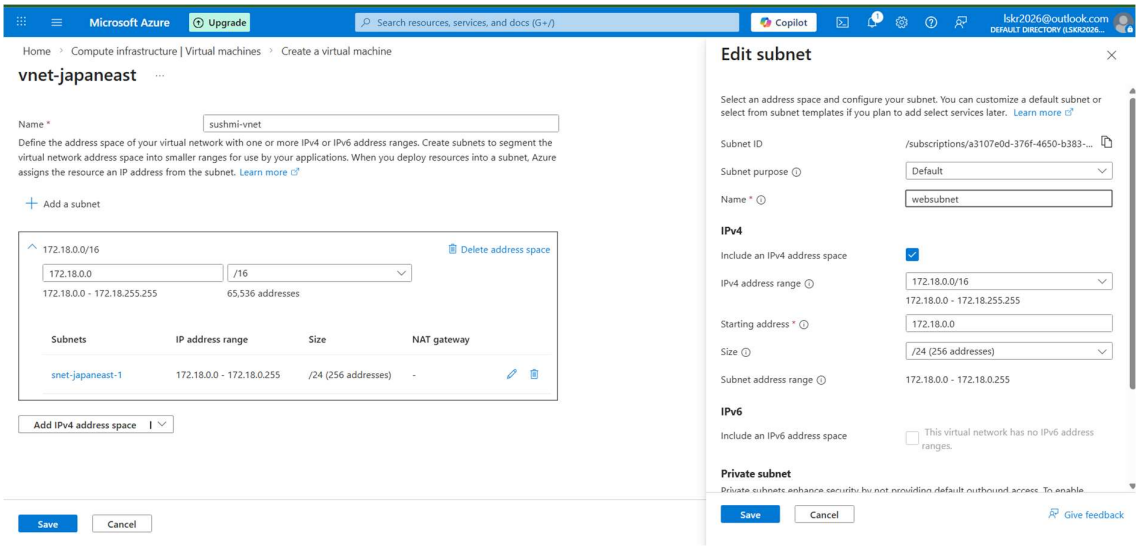
3 -TIER-ARCHITECTURE







vnet-japaneast

Name * sushmi-vnet

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. Learn more

+ Add a subnet

172.18.0.0/16 — Delete address space

172.18.0.0 | /16

172.18.0.0 - 172.18.255.255 — 65,536 addresses

Subnets	IP address range	Size	NAT gateway
websubnet	172.18.0.0 - 172.18.0.255	/24 (256 addresses)	-
appsubnet	172.18.1.0 - 172.18.1.255	/24 (256 addresses)	-
dbsubnet	172.18.2.0 - 172.18.2.255	/24 (256 addresses)	-

Save Cancel

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/) | Copilot | lskr2026@outlook.com DEFAULT DIRECTORY (LSKR2026...

Home > Compute infrastructure | Virtual machines > Create a virtual machine

Create a virtual machine

Help me create a VM optimized for high availability | Help me choose the right VM size for my workload | Help me create a low cost VM

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network: (New) sushmi-vnet (sushmi)
Edit virtual network

Subnet *: (New) websubnet
Edit subnet — 172.18.0.0 - 172.18.0.255 (256 addresses)

Public IP: (new) web-sushmi-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group:
- None
- Basic
- Advanced

Public inbound ports *:
- None
- Allow selected ports

Select inbound ports: Select one or more ports

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

< Previous | Next : Management > | Review + create | Give feedback

https://go.microsoft.com/fwlink/?linkid=2033964&clcid=0x409

Home > Compute infrastructure | Virtual machines > Create a virtual machine

Create a virtual machine

Help me create a VM optimized for high availability | Help me choose the right VM size for my workload | Help me create a low cost VM

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. Learn more about custom data for VMs

Custom data

```
#!/bin/bash
sudo su
apt update
apt install nginx -y
```

Custom data on the selected image will be processed by cloud-init.
Learn more about custom data for VMs

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. Learn more about user data for VMs

Enable user data ☐

Performance (NVMe)

Enable capabilities to enhance the performance of your resources.

Higher remote disk storage performance with NVMe ☐
The selected size is not supported for NVMe. See supported size families

< Previous | Next : Tags > | Review + create | Give feedback

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260127132015 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2...
Subscription: Azure subscription 1
Resource group: sushmi

Start time: 1/27/2026, 1:32:49 PM
Correlation ID: 3c034aab-e1a6-4be8-b531-df7a18ed1d63

Deployment details

Next steps

Setup auto-shutdown Recommended

Monitor VM health, performance and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource | Create another VM

Give feedback

Tell us about your experience with deployment

web-sushmi | Network settings

Virtual machine

List all my network interfaces for this VM +2

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
- Networking
  - Network settings
  - Load balancing
  - Application security groups
  - Network manager
- Settings
- Availability + scale
- Security
- Backup + disaster recovery

Add or remove favorites by pressing Ctrl+Shift+F

Network security group web-sushmi-nsg (attached to networkInterface: web-sushmi198)
Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Prio...	Name	Port	Protocol	Sourc
Inbound port rules (5)				
100	aws	22	TCP	Any
110	htt	80	TCP	Any
65000	AllowVnetInBound	Any	Any	Virtu
65001	AllowAzureLoadBalancerInB...	Any	Any	Azur
65500	DenyAllInBound	Any	Any	Any
Outbound port rules (3)				

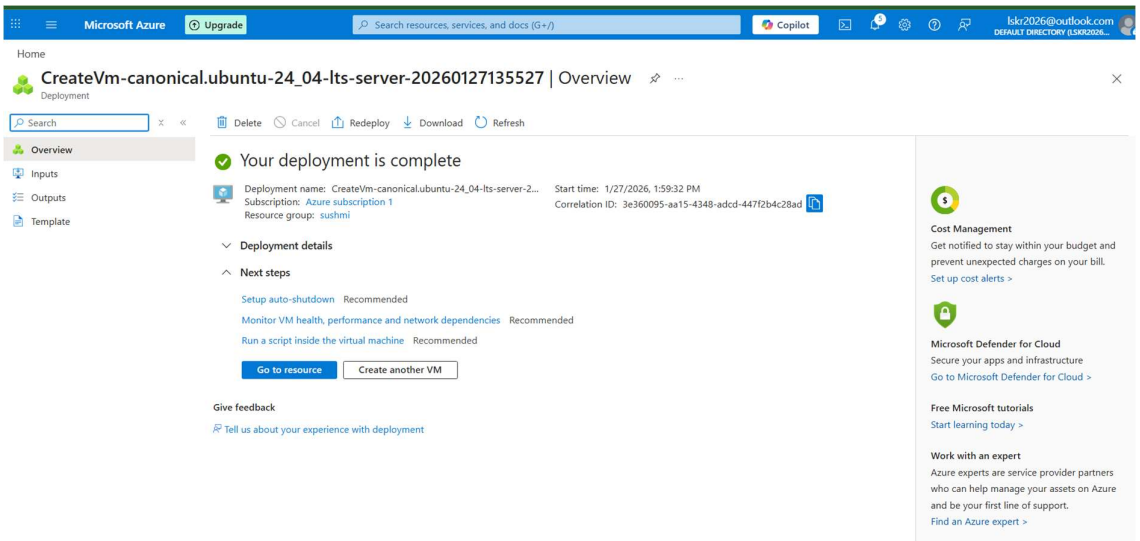
Not secure 20.89.158.169

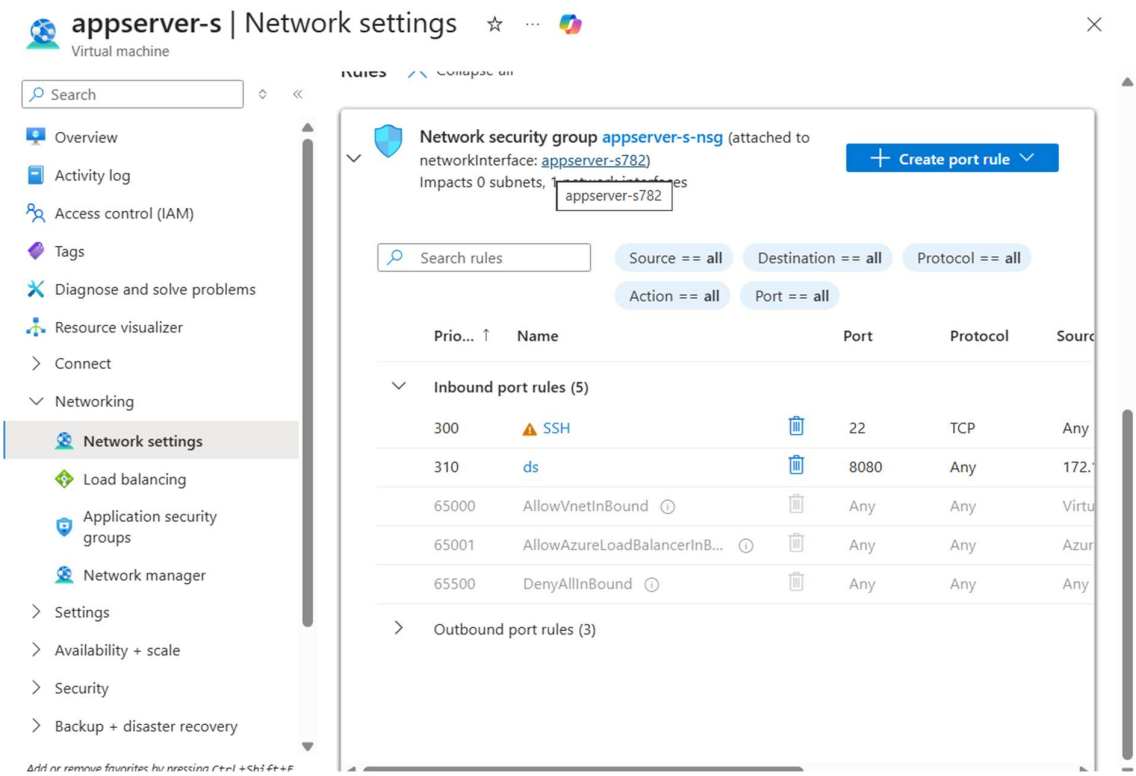
Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

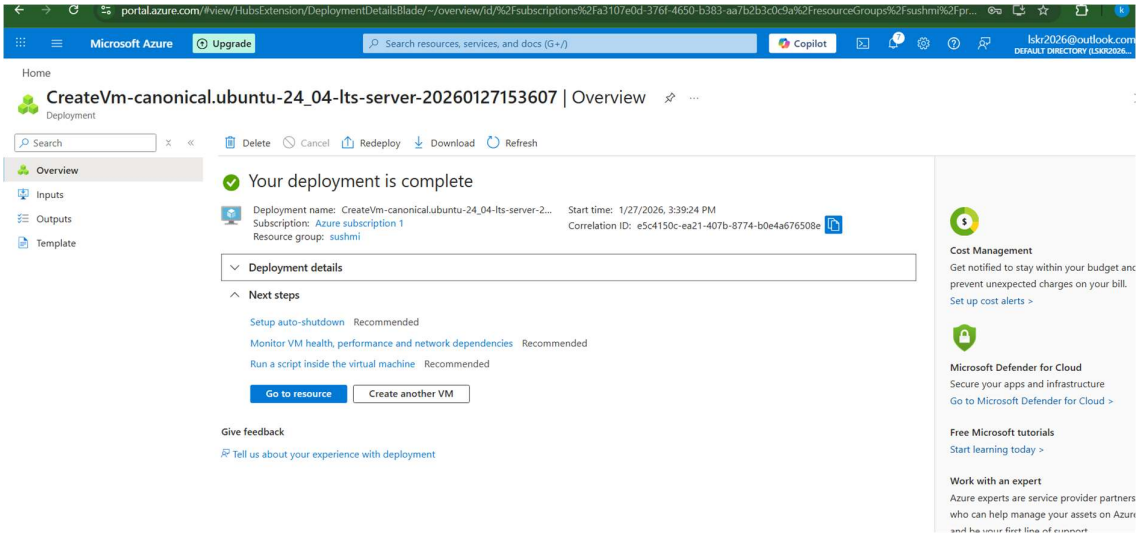
Thank you for using nginx.

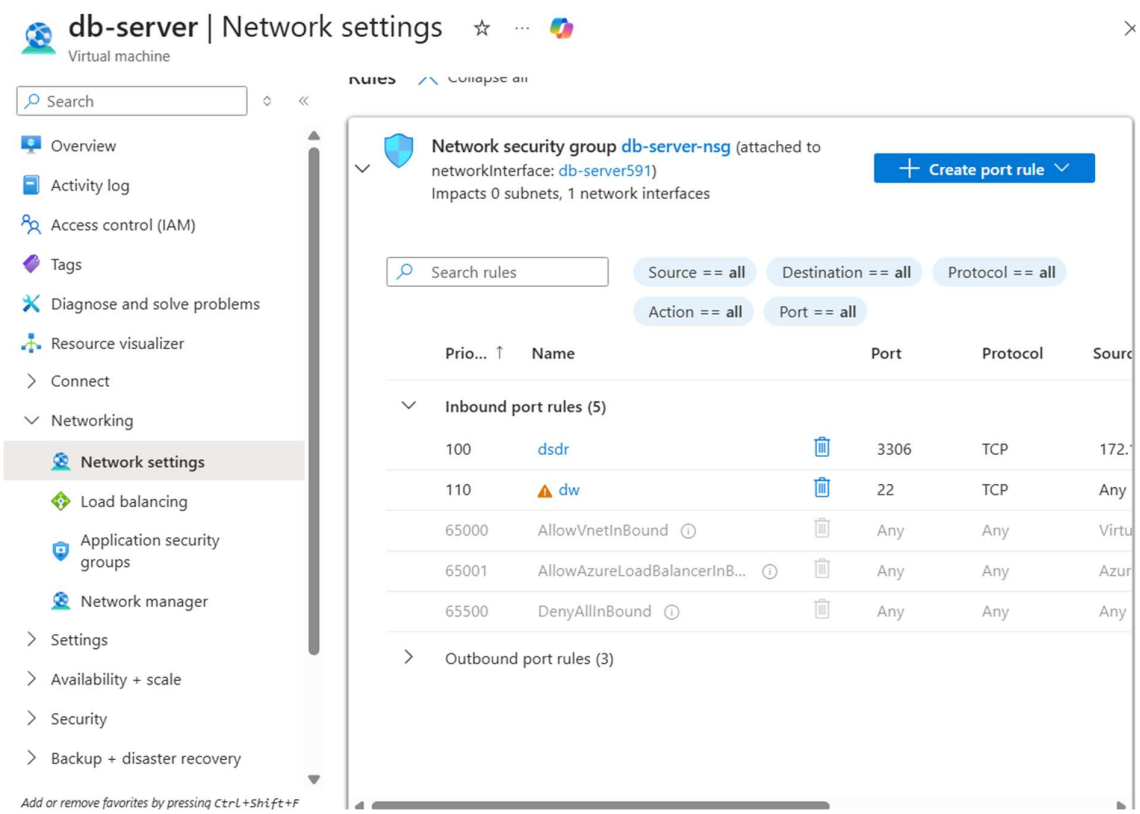


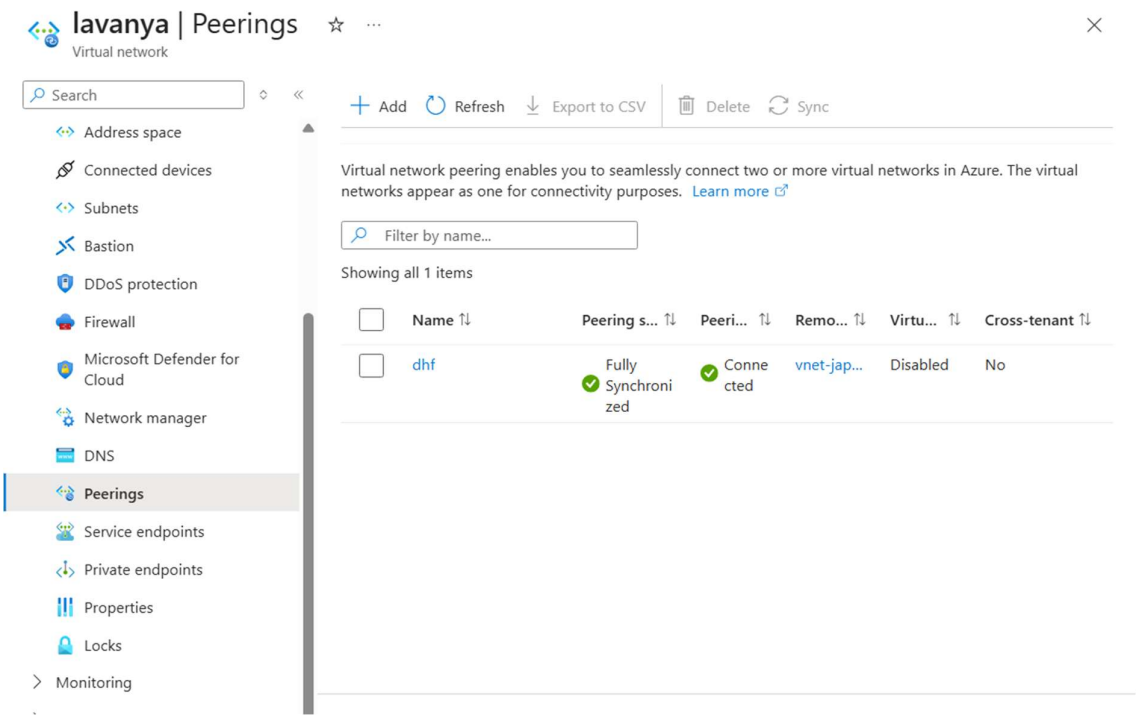


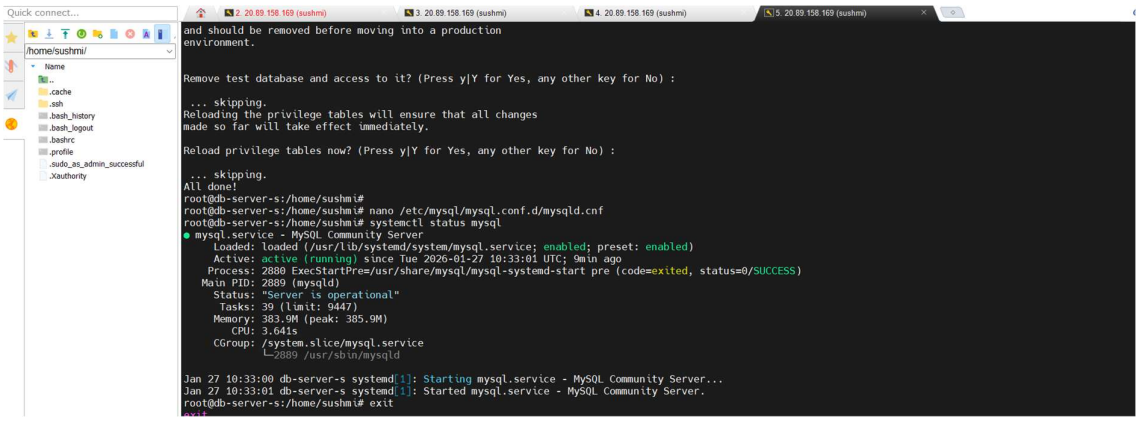

```
root@sushmi-app:/home/sushmi# cd tomcat/bin
root@sushmi-app:/home/sushmi/tomcat/bin# ./startup.sh
Using CATALINA_BASE:   /home/sushmi/tomcat
Using CATALINA_HOME:   /home/sushmi/tomcat
Using CATALINA_TMPDIR: /home/sushmi/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/sushmi/tomcat/bin/bootstrap.jar:/home/sushmi/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@sushmi-app:/home/sushmi/tomcat/bin#
```

```
Connection to 172.18.1.4 closed.
root@web-sushmi:/home/sushmi#
root@web-sushmi:/home/sushmi#
root@web-sushmi:/home/sushmi# exit
exit
sushmi@web-sushmi:~$ telnet 172.18.1.4 8080
Trying 172.18.1.4...
Connected to 172.18.1.4.
Escape character is '^]'.
Connection closed by foreign host.
sushmi@web-sushmi:~$
```






lavanya | Peerings
Virtual network
Search
Address space
Connected devices
Subnets
Bastion
DDoS protection
Firewall
Microsoft Defender for Cloud
Network manager
DNS
Peerings
Service endpoints
Private endpoints
Properties
Locks
Monitoring
Add
Refresh
Export to CSV
Delete
Sync
Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. The virtual networks appear as one for connectivity purposes. Learn more
Filter by name...
Showing all 1 items
Name
Peering s...
Peeri...
Remo...
Virtu...
Cross-tenant
dhf
Fully Synchronized
Connected
vnet-jap...
Disabled
No


Quick connect...
/home/sushmi/
Name
..
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_successful
.Xauthority
2. 20.89.158.169 (sushmi)
3. 20.89.158.169 (sushmi)
4. 20.89.158.169 (sushmi)
5. 20.89.158.169 (sushmi)
and should be removed before moving into a production
environment.
Remove test database and access to it? (Press y|Y for Yes, any other key for No) :
... skipping.
Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.
Reload privilege tables now? (Press y|Y for Yes, any other key for No) :
... skipping.
All done!
root@db-server-s:/home/sushmi#
root@db-server-s:/home/sushmi# nano /etc/mysql/mysql.conf.d/mysqld.cnf
root@db-server-s:/home/sushmi# systemctl status mysql
● mysql.service - MySQL Community Server
Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-01-27 10:33:01 UTC; 9min ago
Process: 2880 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
Main PID: 2889 (mysqld)
Status: "Server is operational"
Tasks: 39 (limit: 9447)
Memory: 383.9M (peak: 385.9M)
CPU: 3.641s
CGroup: /system.slice/mysql.service
└─2889 /usr/sbin/mysqld
Jan 27 10:33:00 db-server-s systemd[1]: Starting mysql.service - MySQL Community Server...
Jan 27 10:33:01 db-server-s systemd[1]: Started mysql.service - MySQL Community Server.
root@db-server-s:/home/sushmi# exit

Quick connect...

6. 20.63.215.35 (sushmi) | 8. 20.89.131.190 (sushmi) | 9. 20.89.131.190 (sushmi)

/home/sushmi/

Name
..
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.Xauthority

```
apache-tomcat-11.0.18/bin/shutdown.sh
apache-tomcat-11.0.18/bin/startup.sh
apache-tomcat-11.0.18/bin/tool-wrapper.sh
apache-tomcat-11.0.18/bin/version.sh
root@appserver-s:/home/sushmi/tomcat# cd /tomcat/bin
bash: cd: /tomcat/bin: No such file or directory
root@appserver-s:/home/sushmi/tomcat# cd tomcat/bin
bash: cd: tomcat/bin: No such file or directory
root@appserver-s:/home/sushmi/tomcat# ls
apache-tomcat-11.0.18  apache-tomcat-11.0.18.tar.gz
root@appserver-s:/home/sushmi/tomcat# cd tomcat/bin
bash: cd: tomcat/bin: No such file or directory
root@appserver-s:/home/sushmi/tomcat# mv apache-tomcat-11.0.18 tomcat
root@appserver-s:/home/sushmi/tomcat# ls
apache-tomcat-11.0.18.tar.gz  tomcat
root@appserver-s:/home/sushmi/tomcat# cd tomcat
root@appserver-s:/home/sushmi/tomcat/tomcat# cd bin
root@appserver-s:/home/sushmi/tomcat/tomcat/bin# ./startup.sh
Using CATALINA_BASE:   /home/sushmi/tomcat/tomcat
Using CATALINA_HOME:   /home/sushmi/tomcat/tomcat
Using CATALINA_TMPDIR: /home/sushmi/tomcat/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/sushmi/tomcat/tomcat/bin/bootstrap.jar:/home/sushmi/tomcat/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@appserver-s:/home/sushmi/tomcat/tomcat/bin# cd ..
root@appserver-s:/home/sushmi/tomcat/tomcat# cd ..
root@appserver-s:/home/sushmi/tomcat# cd ..
root@appserver-s:/home/sushmi# cd ..
root@appserver-s:/home# cd ..
root@appserver-s:/# cd ..
root@appserver-s:/# telnet 172.21.0.4
Trying 172.21.0.4...
telnet: Unable to connect to remote host: Connection refused
root@appserver-s:/# telnet 172.21.0.4 3306
Trying 172.21.0.4...
Connected to 172.21.0.4.
Escape character is '^]'.
```